



Systems Design

Week 2: Mental Models and Cluster Maps
January 27, 2026

Pia Zaragoza

Agenda

- 6:30pm - 8:00pm: Review Readings / Homework
- 8:00pm - 8:10pm: Stretch Break
- 8:10pm - 8:45pm: Lecture
- 8:45pm - 9:20pm: Studio Time

Review Reading

What is a Model?

- If we take the meaning literally, the word model means “a smaller version of something big.”
- Models can be used as tools for comparison, for debate, and to explain various processes such as decision-making, technical details, etc.
- Think about the rudimentary war plans our ancestors made. The chief took a wooden stick and modeled the war field into the sand. He explained where forces should be split, concentrated, a bigger rock they could hide behind, and so on.
- It is much easier to understand and explain things in a visual way.



Thinking in Systems

A system is an interconnected set of elements that is coherently organized in a way that achieves something. Changing relationships usually changes system behavior.

If you look at that definition closely for a minute, you can see that a system must consist of three kinds of things:

- **Elements:** often visible and tangible things
- **Interconnections:** the relationships that hold the elements together
- **Function** (nonhuman) **or Purpose** (human): deduced from behavior and seeing how a system behaves

What are the elements, interconnections, and function / purpose of NMDP?



HOW TO SWAB



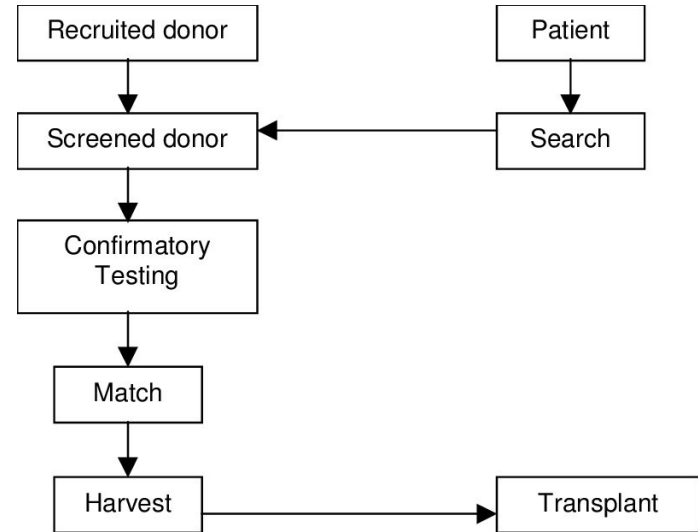
DONATION JOURNEY



Stocks and Flows

- **Stock** is the foundation of any system. It represents the elements that can be seen, felt, counted, or measured at a given moment. In systems terms, a stock is a store or accumulation of material or information that has built up over time. It functions as the system's memory, reflecting the history of changes in its flows.
- **Flows** are filling and draining, births and deaths, purchases and sales, growth and decay, deposits and withdrawals, successes and failures.
- A stock takes time to change, because flows take time to flow.

What are the stock and flows of NMDP?



Stocks

Registered Donor Pool

- People currently listed and eligible to donate
- Primary enabling stock of the system

Available Donors

- Subset of registered donors who are reachable, healthy, and willing at time of request

Patients Awaiting Match

- Patients actively searching for a compatible donor

Matched Donor

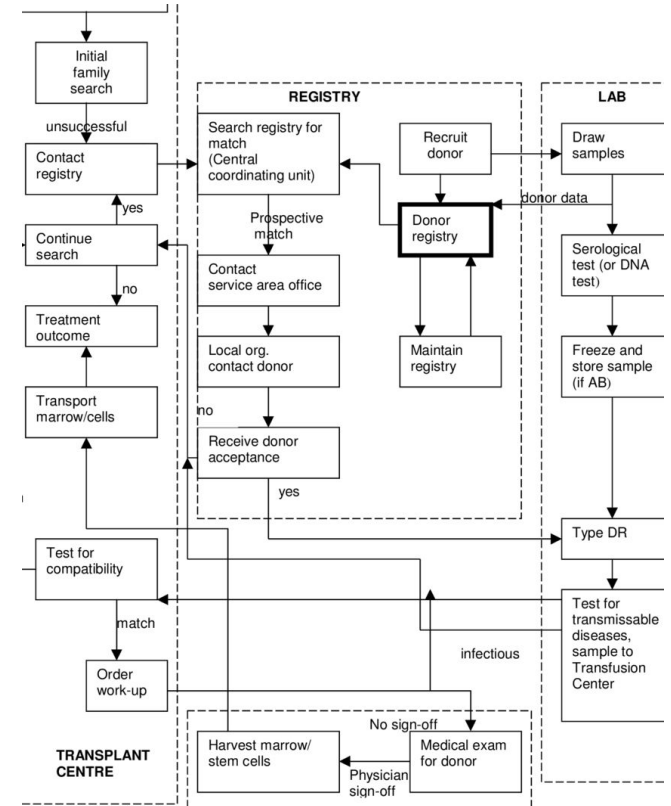
- Temporary holding stock between match and transplant

Completed Transplants

- Transplants successfully carried out

Post-Transplant Survivors

- Patients who survive and recover after transplant



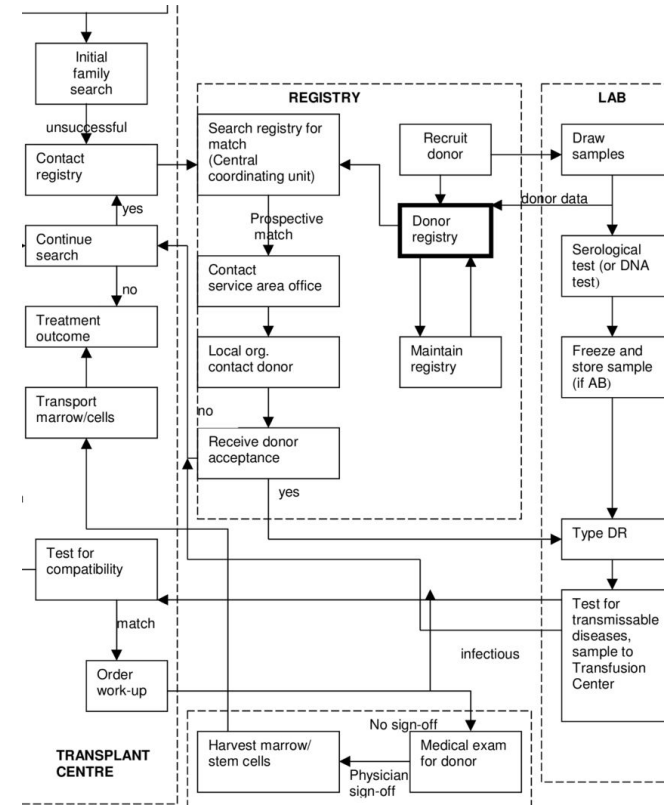
Flows

Donor Enrollment Rate

- Outreach, recruitment drives, UX of sign-up

Donor Attrition Rate

- Aging out, health changes, withdrawal, loss of contact



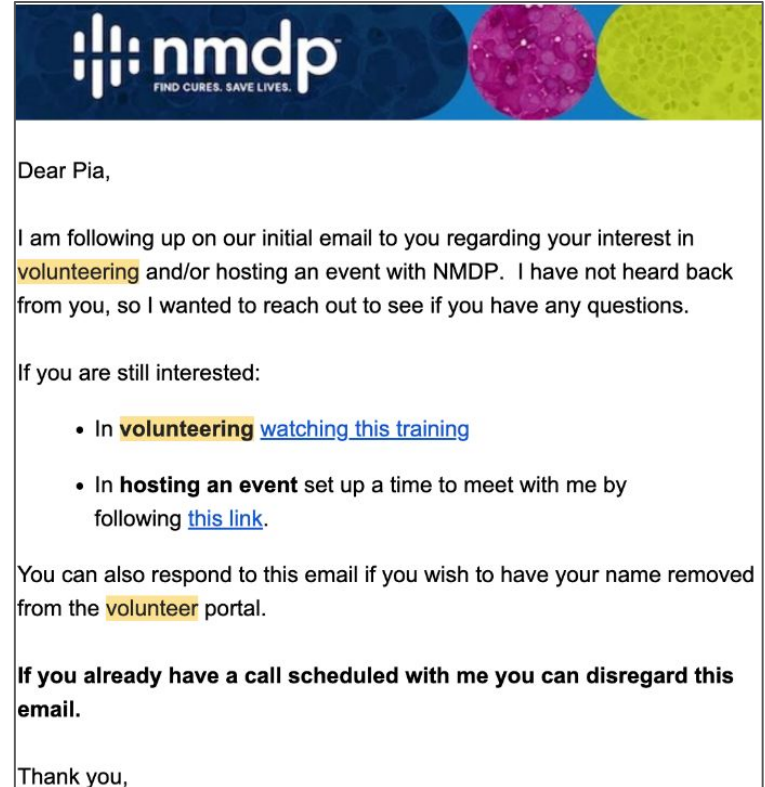
Feedback Loop

Feedback loops are formed when changes in a stock affect the flows into or out of that same stock.

In NMDP, feedback loops adjust donor recruitment, retention, and matching rates in response to patient demand and registry size.

Example

- As the amount of available cash in your checking account (a stock) decreases, you may decide to work more hours and earn additional income.
- The money entering your bank account is a flow that you can adjust to increase your cash stock to a more desirable level.
- Conversely, if your account balance grows very large, you may choose to work fewer hours, reducing the inflow.
- This feedback loop helps keep your available cash within a range that feels acceptable to you.



Feedback Loop

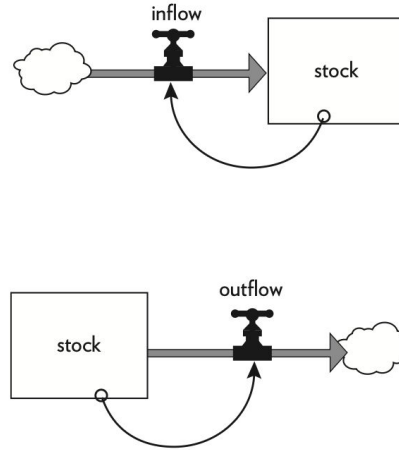


Figure 8. How to read a stock-and-flow diagram with feedback loops. Each diagram distinguishes the stock, the flow that changes the stock, and the information link (shown as a thin, curved arrow) that directs the action. It emphasizes that action or change always proceeds through adjusting flows.

Donor Diversity

- Every year, about 18,000 people in the United States are diagnosed with a life-threatening illness for which a stem cell transplant from a donor is the best treatment option.
- Unfortunately, only about 30% of those patients have a family member who is the best match. That means that about 12,000 people need to find an unrelated donor.
- The process by which the donor and recipient are matched is called [HLA \(human leukocyte antigen\) typing](#). It's not the same as blood type. Instead, it has to do with the immune proteins that we all inherit at birth from both of our parents.
- There is an urgent need for more donors between the ages of 18 and 40, especially donors of non-European and mixed ancestry.

Source: [Sloan Kettering Memorial](#)



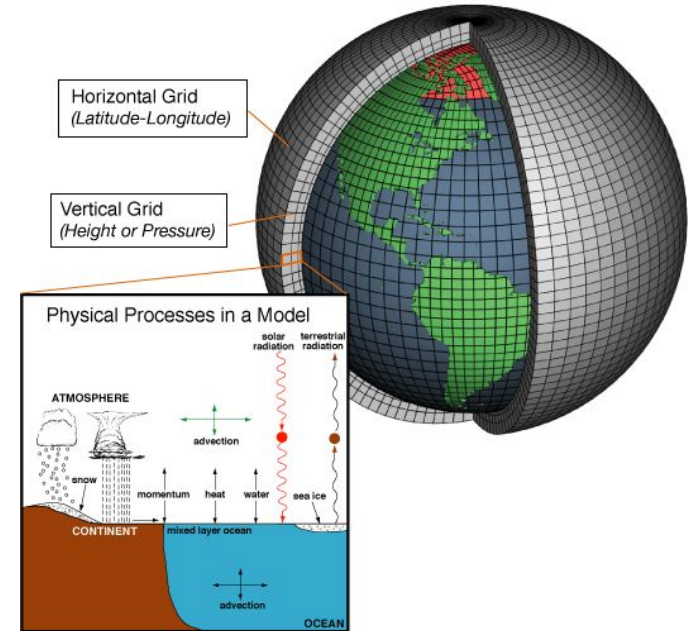
Review Homework

Stretch Break

Models

Key Considerations

- Models are **imperfect and incomplete** imitations of the real world.
- A model won't give you a clear yes or no answer to almost any question
- Models are meant to **paint a broader picture**. They are useful in understanding the underlying causes of patterns.
- To become a great modeler, you have to become very good at asking one question. Namely, why?

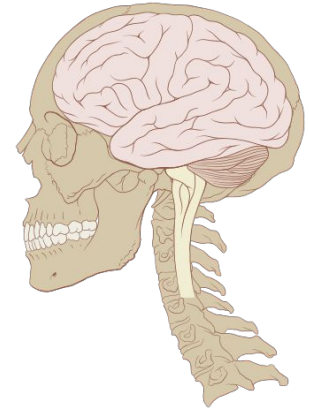
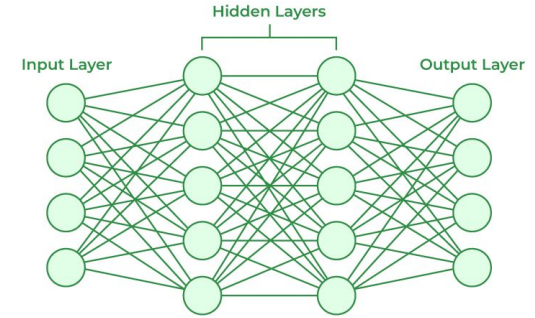


An example of how the Earth is broken down into cubes displaying some of the many different factors climate modelers consider when creating models. -NOAA

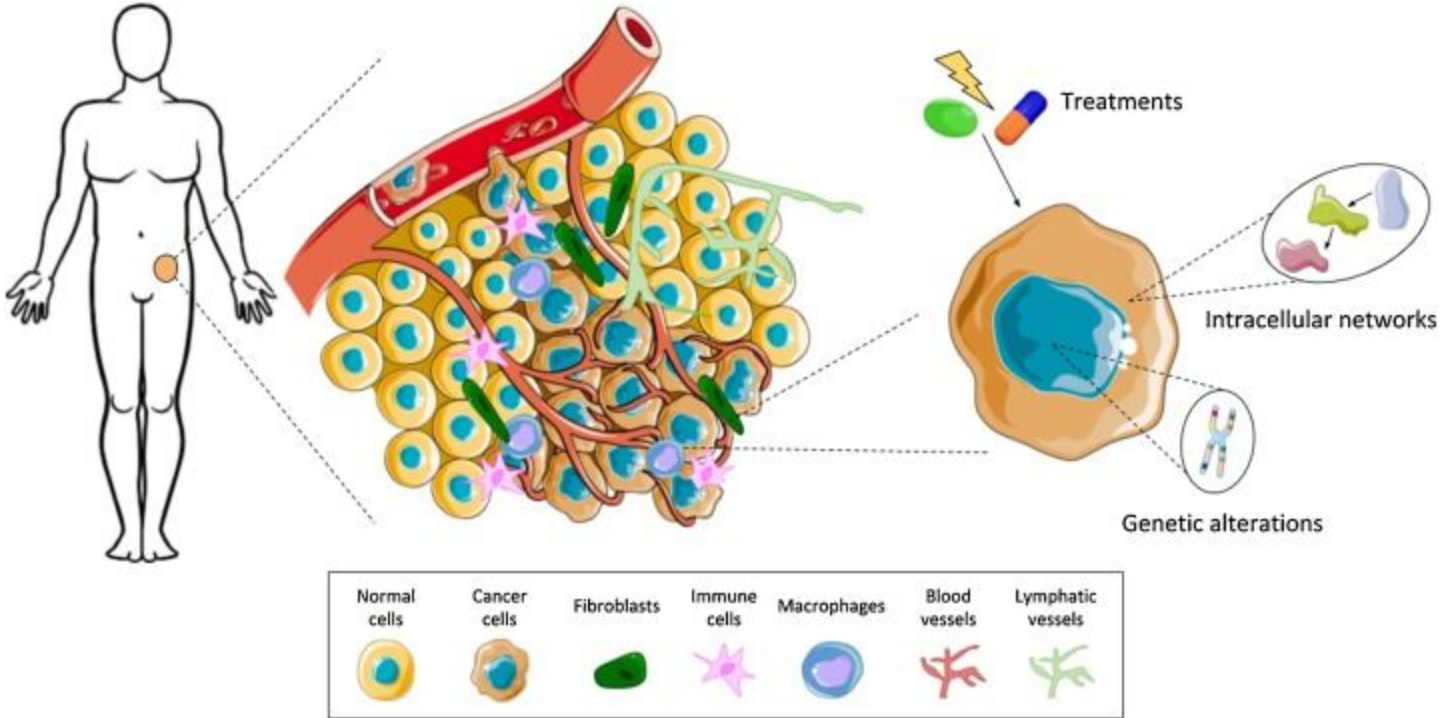
Mathematical Modeling

Mathematical modeling is the art of using mathematical tools and concepts, usually supported by computing power, to represent natural systems, properties and phenomena. Some popular modeling techniques include:

- **Neural network** in computational science is a network of artificial neurons or nodes. The nodes and their relationships are found with the integration of different algorithms that copy the way the brain operates. They identify the best network according to the provided input and output experimental data.
- **Agent-based model** is a class of discrete mathematical and computational models that are able to represent the behavior (actions and interactions) of different autonomous agents and to highlight the whole system as the integration of its different actors.
- **Differential equations** are equations that unknowns are functions and they imply these functions but also their derivative. In the case of dynamic systems, ordinary differential equations (ODEs) are used and the independent variable, or the one in respect to the derivation is performed, is time.

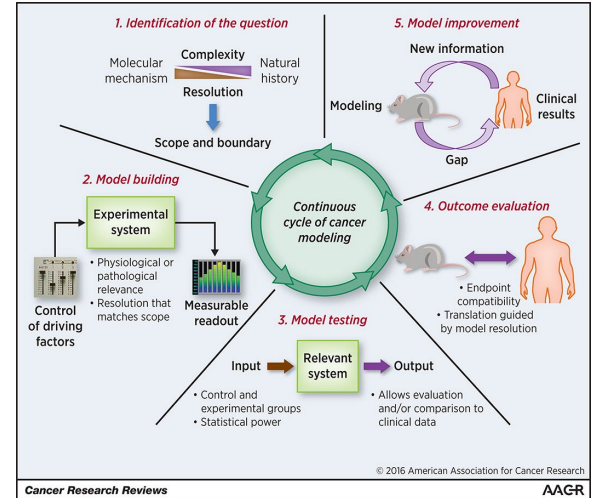


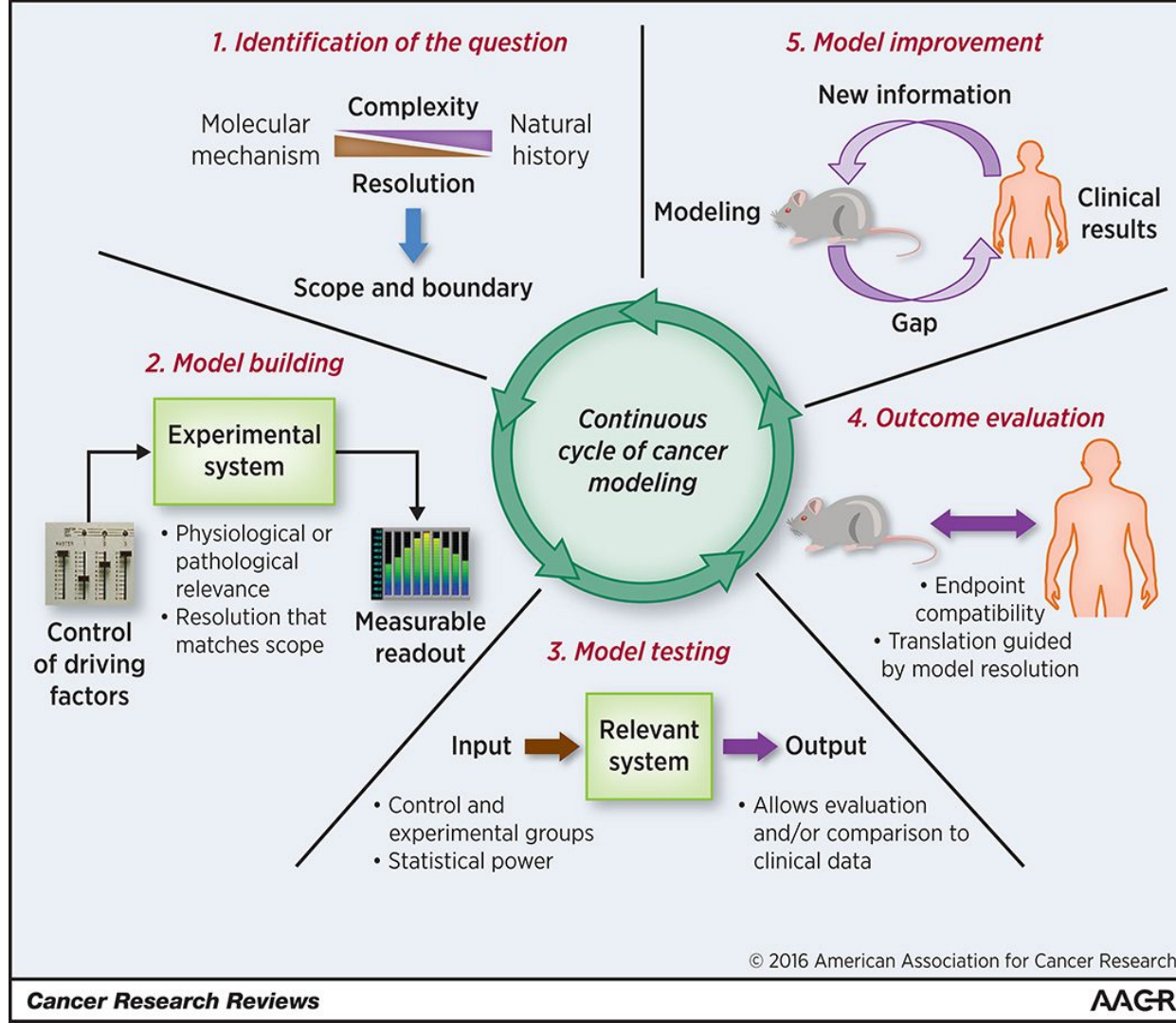
Cancer Modeling: Basic Cancer Research Model



Cancer Modeling

- Cancer is a multiscale and heterogeneous disease, involving a lot of different components and processes.
- This disease really needs strategies able to highlight the interactions between individual components rather than to understand only the individual components.
- Because of the rising amount of generated data and because of their accessibility, computational and mathematical approaches are becoming more and more prevalent to study complex systems of interacting components.
- “Life as an evolving open system can mount an overwhelming number of variables, making it very difficult to identify the boundary of the system and the critical determinants”. -[Concepts in Cancer Modeling](#)





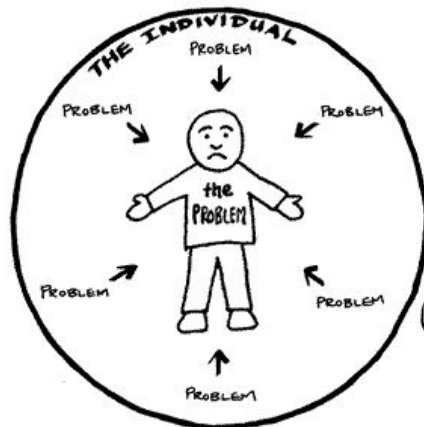
Homework

Mental Models

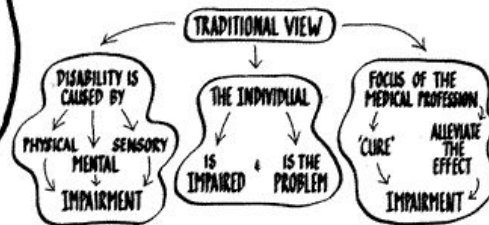
- Mental models are the conscious or unconscious driving force behind our thinking and decision-making.
- Some of the mental models that guide our life we learned from our parents, community, and culture. Some others we adopted by choice. And we rely on them more than we realize when we're trying to **make sense of our surroundings.**
- "Our brain already rehearsed the mental model of traffic lights so frequently that it became an automatic response".



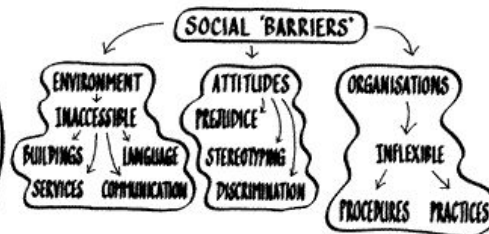
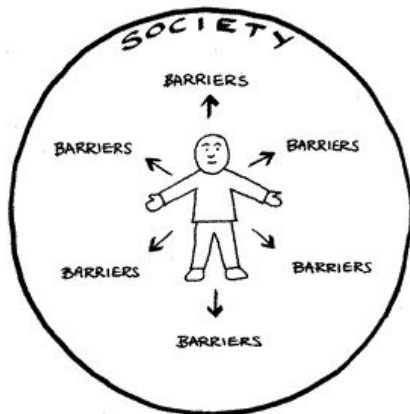
THE MEDICAL MODEL OF DISABILITY



IMPAIRMENTS AND CHRONIC ILLNESS
OFTEN POSE REAL DIFFICULTIES BUT
- THEY ARE NOT THE MAIN PROBLEMS

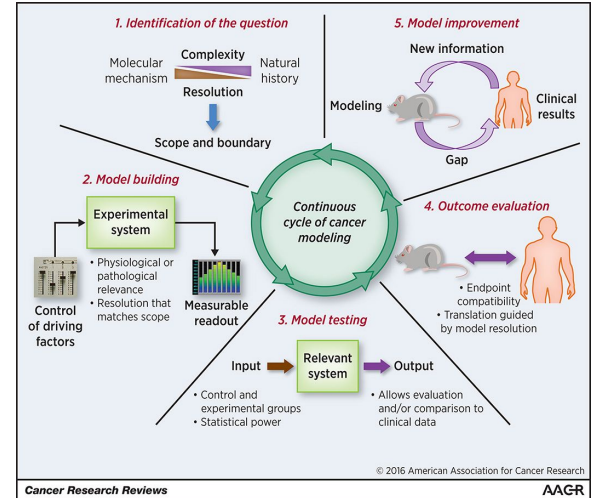


THE SOCIAL MODEL OF DISABILITY



Define

- One should ask a specific question, a central issue the model is meant to explore. In other words, we need to define the boundaries of the problem we want to model within the system. This defined question will be the guidepost for your model.
- The defining question will lead you to draw boundaries around the elements and interconnections that are important to answer the question and to leave out the parts that are not.
- A model without a clearly defined question becomes directionless, cluttered, and unhelpful.
- A good modeler keeps their assumptions in check and knows that both under- and over-generalizing are detrimental.
- “All models are wrong, but some are useful.” -George E.P. Box



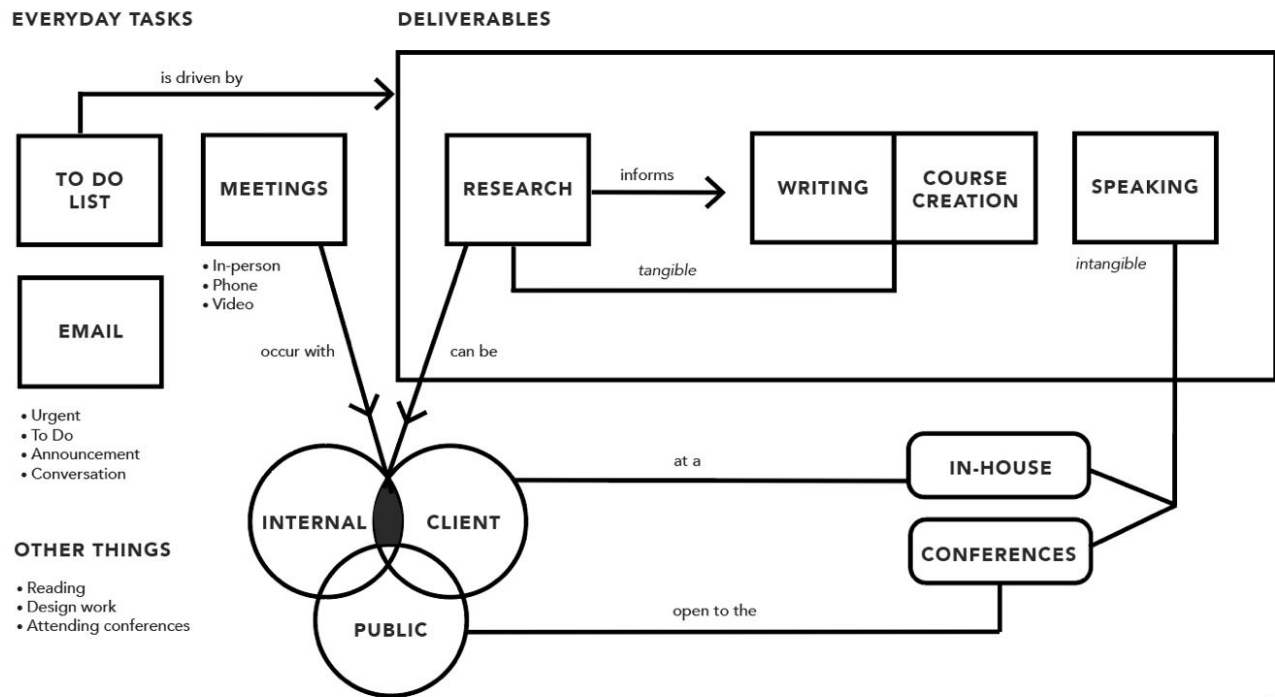
Mental Models

Cognitive maps, mind maps, and concept maps look and feel similar; this similarity causes confusion. They are three different ways of visualizing a mental model.

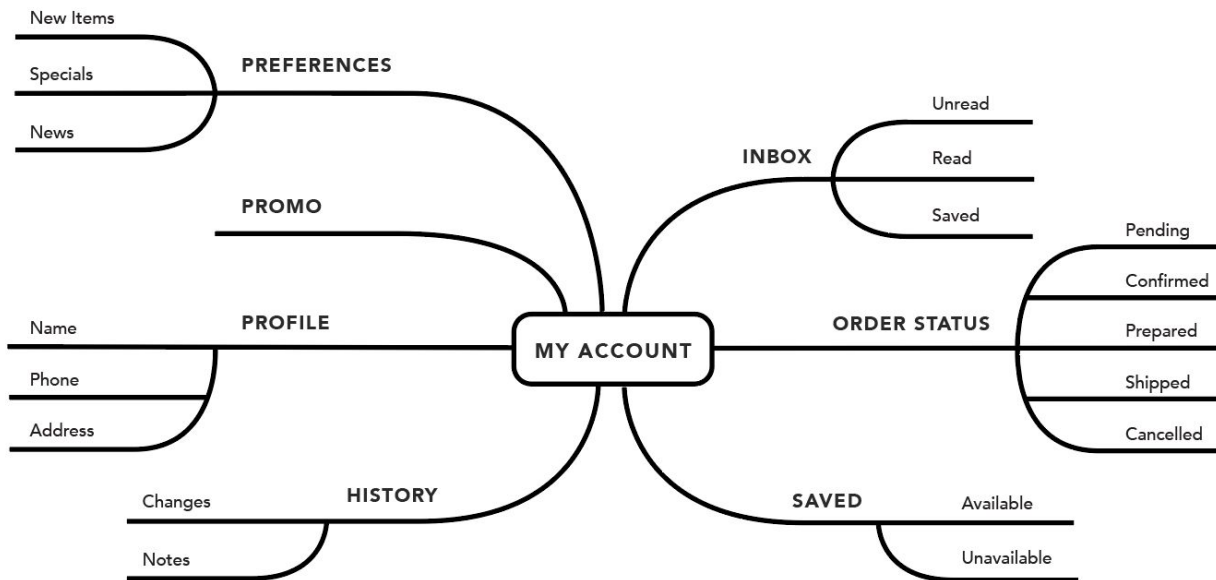
- **Cognitive Maps:** no visual rules that they need to obey.
- **Mind Map** all nodes (except the tree root) have only one parent node. Each node can have children corresponding to that concept's subtopics.
- **Concept Map** a graph in which nodes represent concepts and are related through labeled, directed edges that illustrate relationships between them.



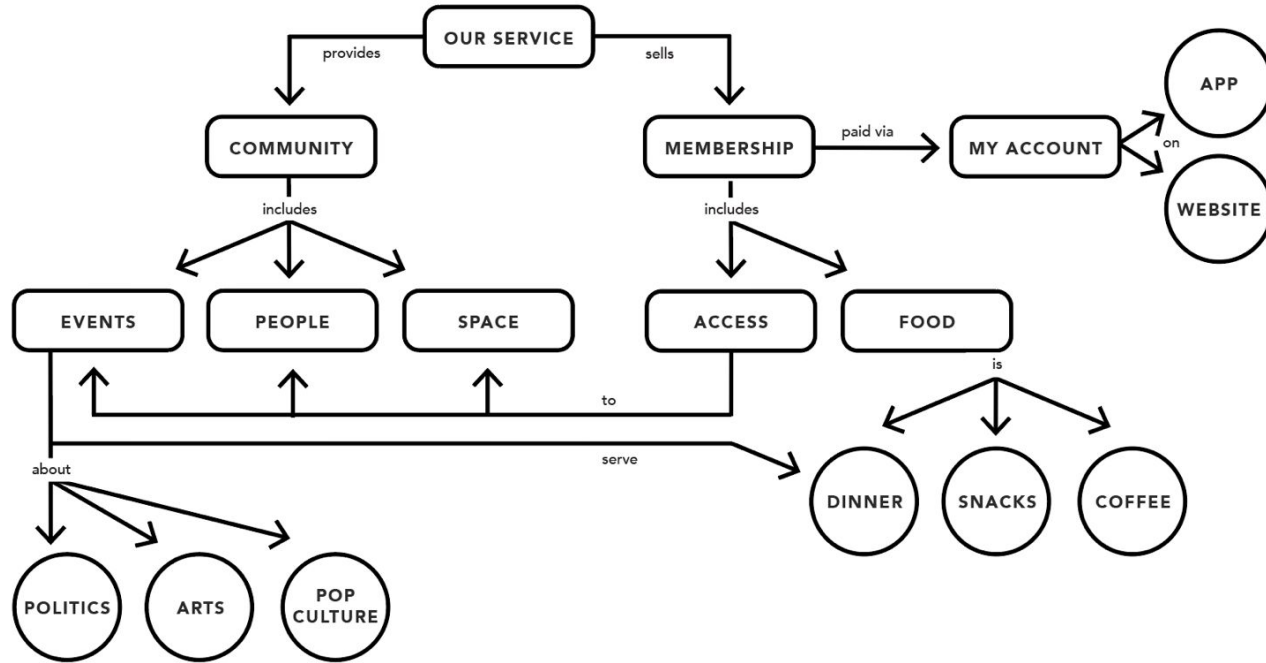
Cognitive Mapping



Mind Map

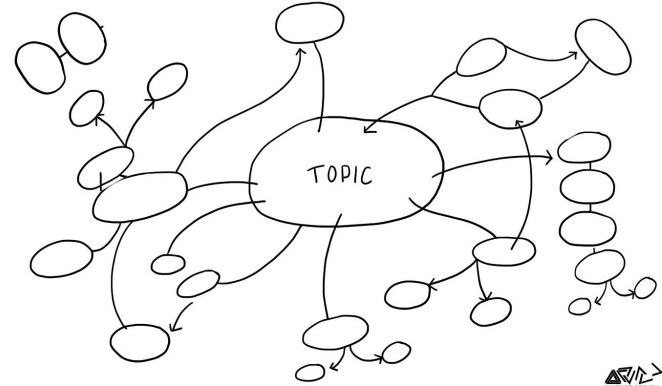


Concept Map



Systems Mapping

- Systems mapping is a tool to help you **visualize the system** you are analyzing. It requires an understanding of the elements, their functions, and their interrelationships within the system.
- It is essential to pin down and accurately **assess relevant leverage points** where our interference would bring the greatest benefits.
- Commonly used methods include: cluster maps, interconnected circle maps, causal loop diagrams, behavior-over-time graphs, and stock-and-flow diagrams.



Thank you!
